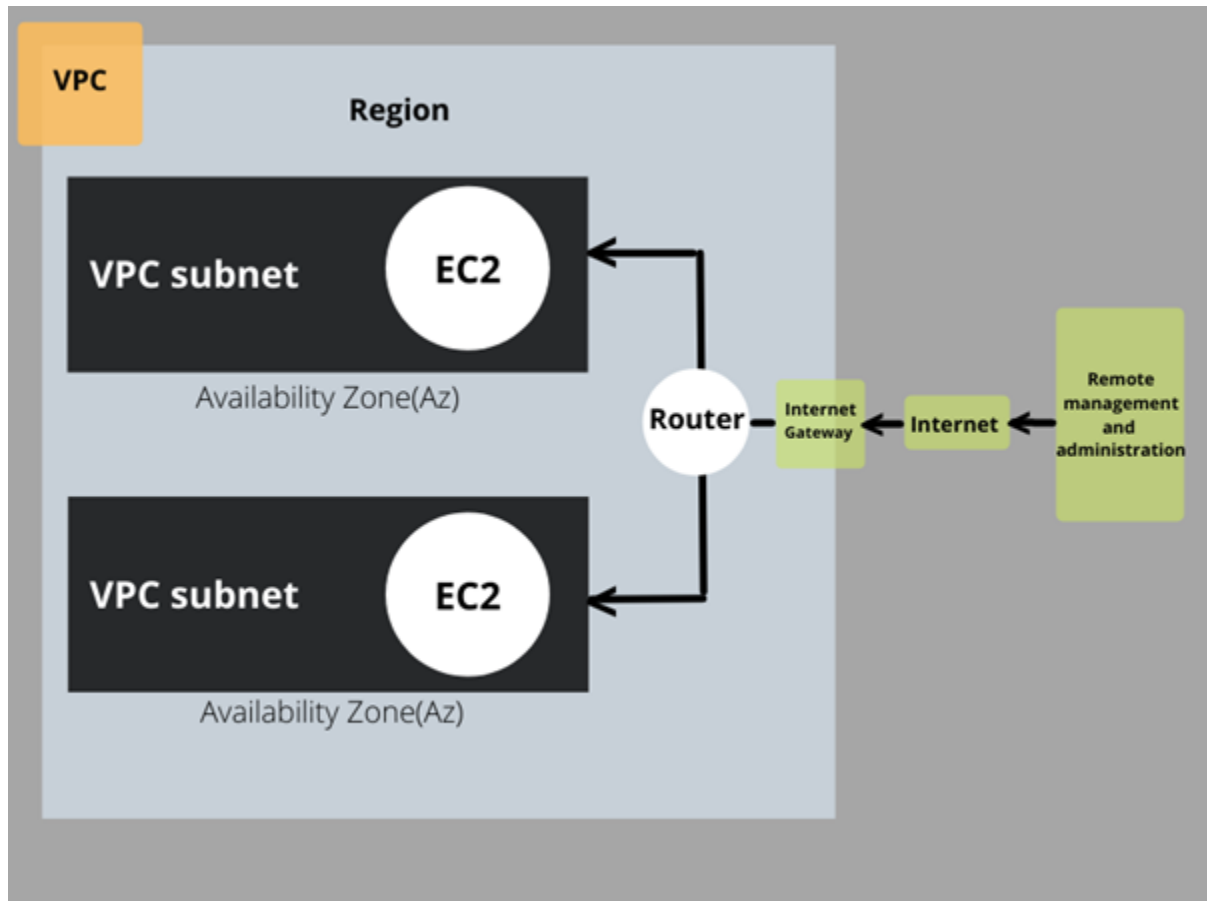


Core Components: Inside the VPC

These are the building blocks you use to construct your private network.



1. Subnets (Public vs. Private)

A subnet is a range of IP addresses in your VPC. You launch resources (like EC2 instances) inside a subnet

Public Subnet: Has a direct route to an Internet Gateway. Resources here have public IP addresses and can talk to the internet directly. Used for: Load Balancers, Bastion Hosts, NAT Gateways.

Private Subnet: Has NO route to an Internet Gateway. Resources here have only private IPs. They cannot accept traffic from the internet. Used for: Application Servers, Databases.

2. Route Tables

A set of rules, called routes, that determine where network traffic from your subnet or gateway is directed.

Every subnet must be associated with a route table.

Local Route: Every route table automatically gets a "local" route (e.g., 10.0.0.0/16 -> Local), which allows all resources inside the VPC to talk to each other.

3. Internet Gateway (IGW)

A horizontally scaled, redundant, and highly available VPC component that allows communication between your VPC and the internet.

- It performs Network Address Translation (NAT) for instances with public IPv4 addresses.
- You attach one IGW to a VPC.

4. NAT Gateway (Network Address Translation)

A managed service that allows instances in a private subnet to connect to the internet (e.g., for software updates) but prevents the internet from initiating a connection to those instances.

- **Placement:** You must create the NAT Gateway in a Public Subnet.
- **Routing:** You update the Private Subnet's route table to send internet traffic (0.0.0.0/0) to the NAT Gateway.

5. Security Groups & Network ACLs (Firewalls)

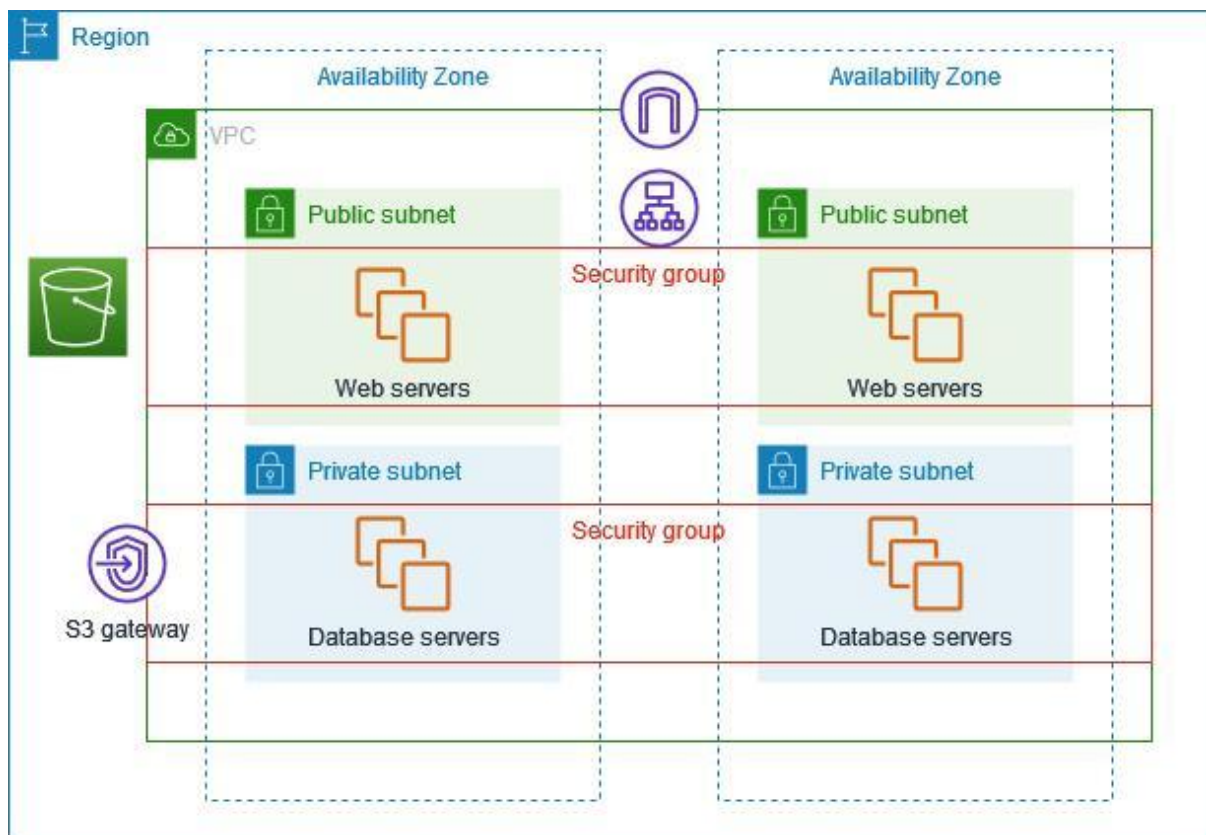
- **Security Groups:** Acts as a virtual firewall for your instance to control inbound and outbound traffic. It is stateful (if you allow a request in, the response is automatically allowed out).
- **Network ACLs (NACLs):** An optional layer of security for your subnet. It acts as a firewall for controlling traffic in and out of one or more subnets. It is stateless (you must explicitly allow traffic in both directions).

Different VPC scenarios for hosting enterprise applications

1. Single VPC with Public/Private Subnets (2-Tier Architecture)

This is the standard approach for simple, web-facing applications requiring high availability.

- **Structure:** One VPC divided into public subnets (for load balancers) and private subnets (for application servers and databases), spanning multiple Availability Zones (AZs).
- **Pros:** Easy to set up, minimal network management, cost-effective, high availability.
- **Cons:** Limited scaling, potential for resource exhaustion as the enterprise grows, lack of network isolation between different application environments.
- **Best For:** Development/testing, small-scale web applications, and standalone internal tools.



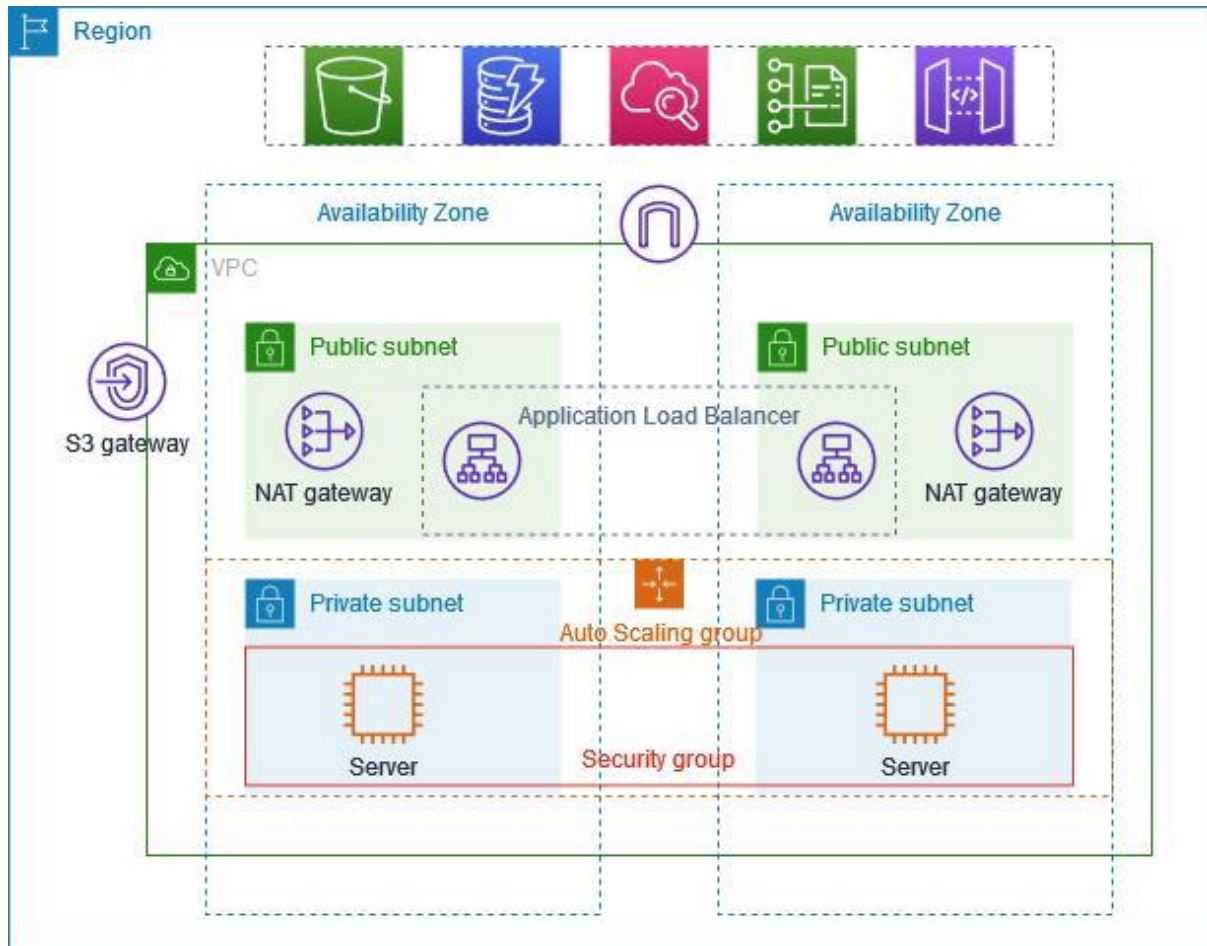
Advantages

1. Simple to set up
2. Low cost
3. Good for testing or small apps

Disadvantages

1. No security isolation
2. Database exposed to internet
3. Not suitable for enterprise

2. Two-Tier VPC (Web + Database)



1. Web layer in **public subnet**
2. Database in **private subnet**

Components

Public subnet (Web servers)

Private subnet (DB servers)

NAT Gateway for outbound internet

Security Groups

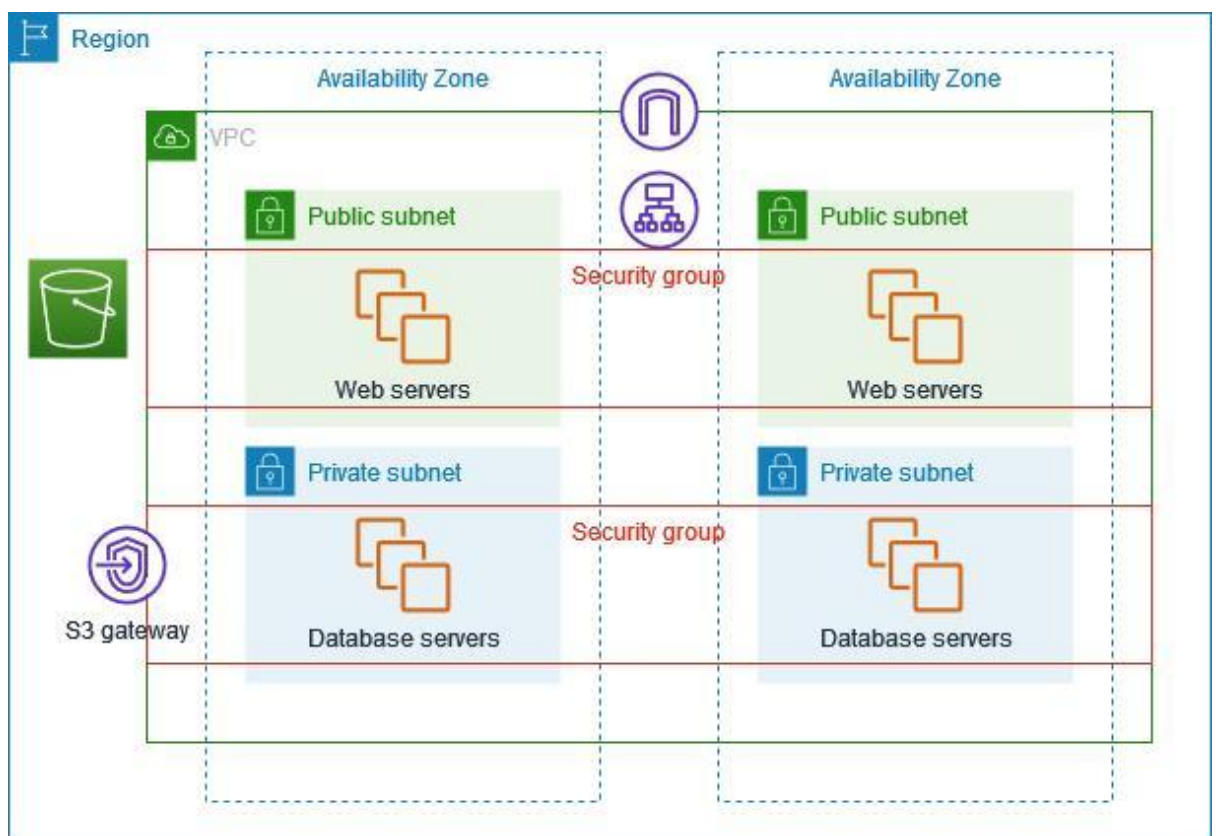
Advantages

- Improved security (DB not public)
- Separation of layers
- Common real-world setup

Disadvantages

- Limited scalability
- App logic mixed with web layer

3. Three-Tier VPC (Web + App + Database)



1. Web layer → Public subnet
2. App layer → Private subnet
3. DB layer → Private subnet

Components

1. Load Balancer
2. Public subnet (Web tier)
3. Private subnets (App + DB tiers)
4. NAT Gateway
5. Auto Scaling

Advantages

1. Strong security isolation
2. Highly scalable
3. Industry standard for enterprise

Disadvantages

1. More complex
2. Higher cost